ETHICAL HACKING

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1) Use Google and whois for Reconnaissance

* Practical 1(A) *

① Site~~colon~~ (:) colon:-
eg:- site : wikipedia.org artificial intelligence
keyword

② filetype colon (:):-
eg:- ① filetype : pdf budget 2024
keyword
② filetyp: doc/png/jpg etc.

③ eg:- link contain wikipedia.org. (Give all related that contain such type of link)
eg:- link: wikipedia.org
keyword

④ eg:- cache: example.com (to find cache)
keyword

⑤ eg:- intitle: seo (title contain that keyword)
keyword

⑥ eg:- inurl: machine learning (url/link ~~that~~ contains that keyword)
keyword

⑦ eg:- site: wikipedia.org filetype: pdf
keyword (pdf of that particular keyword)

11) site: facebook.com filetype: pdf machine-learning
(machine learning thing pdfs in face book site)

SSL https -> secure
http -> not secure

(8) ~~eg~~ google.com/advanced_search (provede more functionality to search more advancely)
- can be used when user don't know the formal way of searching particular site or urls etc.

* Practical 1(c)*

① who.is [all domain info]
ipaddress

② https://mxtoolbox.com/DNSlookup.aspx
[domain info]

③ https://toolbox.googleapps.com/apps/dig/
next step -> www.pugcst.in (search it)

[Prac - 2] 2] use CryptTool to encrypt/decrypt passwords using RC4 algorithm & use Cain and Abel for cracking windows account password.

* Open ~~Crpto~~ Cryptool → file new

Encryption/Decryption ← write text (eg:- Hello world)

→ Symmetric (modern

Select ~~Encrypt~~ RC4

key bits (eg:- 16)

→ write key {eg:- ABCD ...} encryption

remember it where encryption happens (select same page)

Encryption/decryption ← ~~dec~~

→ Decryption → Symmetric/modern

~~Decrypt~~ ← RC4

→ Select same keybits → write same text [eg:- ABCD]

Decrypt → Show the same text what we written before. (eg:- Hello world)

Practical (3) (P-3)

[3. Using Traceroute, Ping, ifconfig, Netstat command]

In

* → windows → tracert
  Linux → traceroot

→ :- To find the path taken by packets from your computer to a destination server.

eg:-

→ tracert www.google.com
destination.

→ tracert www.pvgcst.in

* → windows → ping
  Linux → ping

:- To check weather a host is reacheable and measure response time.

eg:-

→ ping www.google.com

ping www.pvgcst.in
192.168.1.1
ping ~~10.255.255.1~~ → o/p → Time out (means private) not ~~reachable~~ reachable.

cls → To clear.

* → windows → ~~ifconfig~~ ipconfig
Linux → ~~ipconfig~~ ifconfig
:- To view IP address & network configuration

* → ~~net~~ windows + linux → netstat
:- This is to view display active network connection & listening ports.

* → windows + linux → arp ~~-a~~
:- To show IP to which shows to MAC address mapping
eg:-
arp -a

* → windows + linux → nslookup
:- To find IP address of website & ~~ver~~ verify DNS is working
(Domain name system)
eg:-
nslookup www.google.com
nslookup www.pvgcst.in

* → windows + linux → route print
:- It is use to display routing table.

* → window + Linux :- getmac
:- To display mac-address.

* → window + Linux :- hostname
:- To display hostname

[4. Using Nmap scanner to perform port scanning]

① ~~(ACK, SYN, FIN, NULL, XMAS)~~ [4th practical]

[N-map] (ACK, SYN, FIN, NULL, XMAS)

cmd → nmap → Enter → If command not found → ~~not have~~ → you don't have

(do if not have)

OR

If showing data → means already have.

[nmap.org | download # windows] search & download from chrome

On cmd search

→ ~~nmap -scan~~

→ nmap scanme.nmap.org [will show the open ports for safe site]

search

→ nmap pvgcst.in [open port we can hack it]

search

→ nmap -f scanme.nmap.org → [because of -f show the starting Top 100 ports only]

o/p → 97 closed port & 3 are open ports showing

search

→ nmap -sn scanme.nmap.org [for ping scan]

→ nmap -p 22,80 scanme.nmap.org [port scan]

②

→ search
→ nmap -sA scanme.nmap.org
[acknowledgement scan]

→ search
→ nmap -sS scanme.nmap.org
[stealah scan]

search
→ nmap -sF scanme.nmap.org
[finish-fin]

→ search
→ nmap -sN scanme.nmap.org
[nullscan]

search
→ nmap -sX scanme.nmap.org
[christmas or X-mas scan]

5.

[Use wireshark sniffer to capture network traffic and analyze]

[Practical 5]

Wireshark → Ethernet (see this) (1st on it then follow steps)

Google chrome → http://httpbin.org

fill form ~~fill forms~~ ← Html forms (Lower side.)

→ submit → open wireshark (before only)

POST (send) ← http (search)

→ right click → follow → TCP stream & https stream

repeat ← If not (seen) ← [search now filled data.]

Then also not coming then use Mozilla (firefox) → fill form then

all steps. ← repeat again. &

(6. Simulate Persistent Cross Site Scripting (XSS) Attack)

(p-~~8~~6)

(Cross Scripting) (Starting steps are same as p-8)

Oracle virtual Box → Windows 7

Google chrome ← xamp on → MySQL → Apache

↳ Search → localhost/dvwa

login ← ~~admin~~ → User id → admin, password → password

↳ dvwa security (left side) → Set it from Impossible to Low

It like a form (normal form) Name* = test ← XSS Stored (left side) ←

Message* = ~~<script>~~ Test message.

↓ (to infect Javascript here, it is dangerous)

Name *

Message * = <script> alert("This is exploit") ~~</script>~~ </script>

(It will execute) ← Click on Sign Guestbook

① 7. Session Impersonation using firefox and Tamper Data add-on)

Ethical Hacking (Practical-7)

[Tamper data]

firefox → Extensions & themes → search at (find more add-ons) → tamper data

Right click on [ ] ← Tamper data for ff Quantum by Pamblam (add)

↓

pin to overflow menu

↓

Now search at main search box [http://httpbin.org] → scroll it other utilities • HTML form ← click

click on [ ] ← (Top right near [ ]) ← submit ← fill the form

→ list Type | Description → click on Yes (start-Tamper Data?)

← new page open data can be change according to us ← click OK [on Extension: Tamper Data for...] ← again fill the form submit it

② (Tamper data)

→ Click on [Add Header] → Add this [X-forwarded-for | 192.168.2.14]

add new info such as X-forwarded-for

↓

click on OK

↓

will added on origin : 192.168.2.14

In short

Go-to-link → http://httpbin.org → fill form but cannot change

On it (tamperdata) → Goto link → fill form

↓

Submit ⇐ box will come click (OK) ← Add header

↓

[X-forwarded-for | 192.168.2.14]

↓

OK ← Add on origin

(P→8) 8. Perform SQL injection Attack (SQL Injection)

→ Oracle Virtual Box → Windows 7

Start (loads) take little bit time → Open Browser → Comeback to ~~Oracle~~ Window 7

↳ Click show → Starting windows

password is admin → ~~Google chrome~~

↳ xampp → Google Chrome → Search
→ on MySQL
→ on Apache

localhost/dvwa → user → admin
password → password

↳ login → Dvwa security leftside

Impossible set to Low

submit → SQL injection leftside

↳ check by → 1
→ 2
→ 1' or '1' = '1

(not ending with single quotes (' ') because it is already having (putting by own) hence)

(Prac-9) Keypress

(9. Create a simple keylogger using python)

cd Documents
folder
python keylogger.py
filename

Virtual box → windos open → ~~Documents~~

cmd ← python code
→ cd Documents
→ python keylogger.py

Save → named Keylogger.py
location → Documents.

→ show run error → do anything logen or in notepad

again. go to Documents ← close it

→ Open txt files of same name keylogger.py → Everything well appear

~~script~~